Day 1

2 June
Data Analytics
Week 1
Data Foundations Platforms

* Data Brics : Coding - SQL
Links on additional resources: - Machine learn
Youtube: Alex the analyst
Certificate. - Microsoft. - Python
Azure fundamental

* GitHub - Portfolio (Repository)
store code, presentation

* Canva
* Data Presentation
* Data Studio
* Lovable.dev
* GCP
* Google Sheets - Excel
* SQL
* Miro.com.

Draw a
app.diagrams.net

3 June

Data:
a collection of facts. Its information can be used to answer business question. Data is everywhere, comes in different formats

Structured data
- refers to data stored in a table with rows + columns
- consistent format

ID	col 1	col 2	col 3
11	row 1		
121	row 2		
12	row 3		

- Able to use to extract insights
  - excel file (spreadsheet)
  - membership form

Unstructured data
- no specific format
- inconsistent format
  - picture
  - podcast
  - social media post
  - video
  - pdf document
  - website

Unique Key
company number
PPS

(data collection) → data driven decisions ← Database

Data driven decision

Decisions made base of facts seen in the past. Ensuring customer understand - to better grow the business

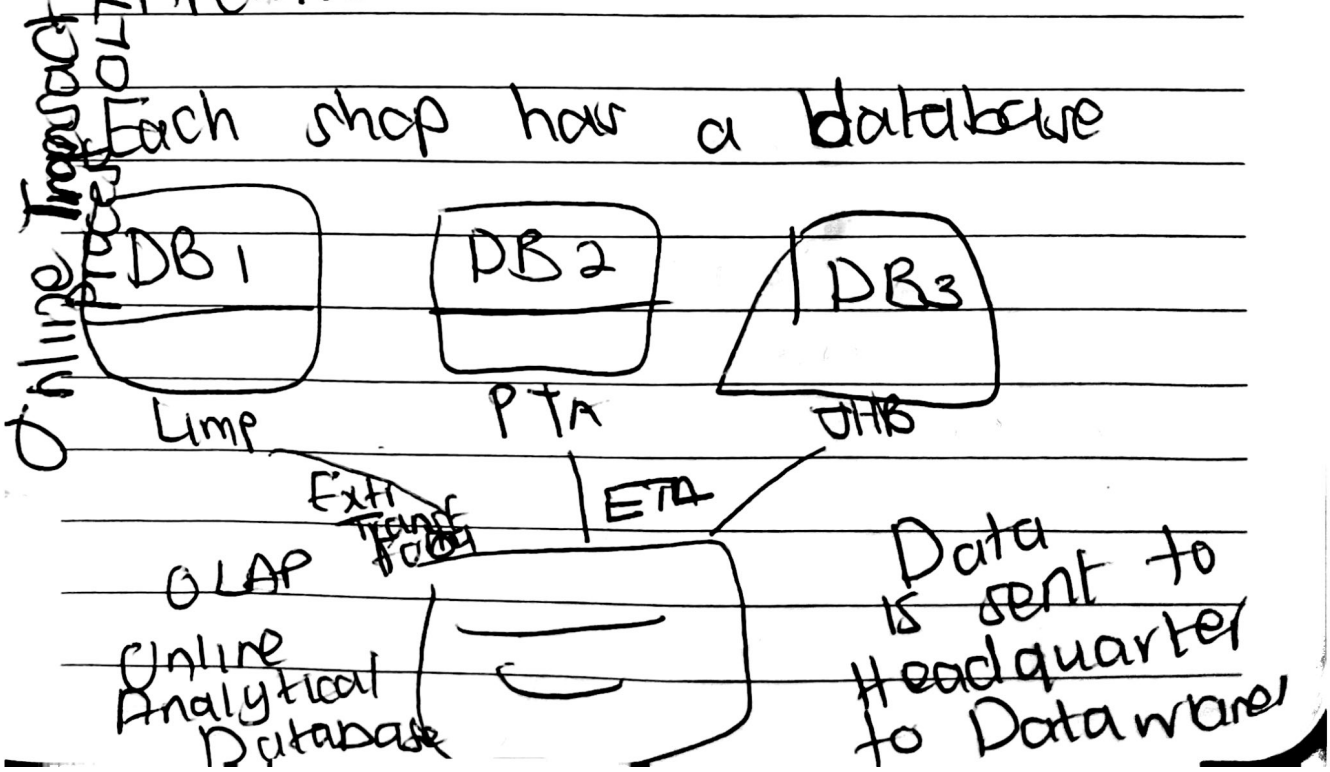
* Data is linked to time
  eg. Shoprite closing @ 10pm
  Mac d's deciding 24/7

* Can be distinguished using an ID

4 June

Data flow / Ware housing Architecture.

Each shop has a database



OLTP - real time data base that store information collect on that point operating hours of a shop

- Its does not allowed It is not used for analysing data. Only used to capture
- Capture as you transact
- Basic SQL Queries (create insert update delete)

OLAP - Data Analyst + Scientist

- Able to analyse to captured data.
- Machine learning
- Processing
- Report.

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ETL Pipeline

A tract from ZZZ to shqprile

- designed by a data engineer

Trasports from the source to warehouse

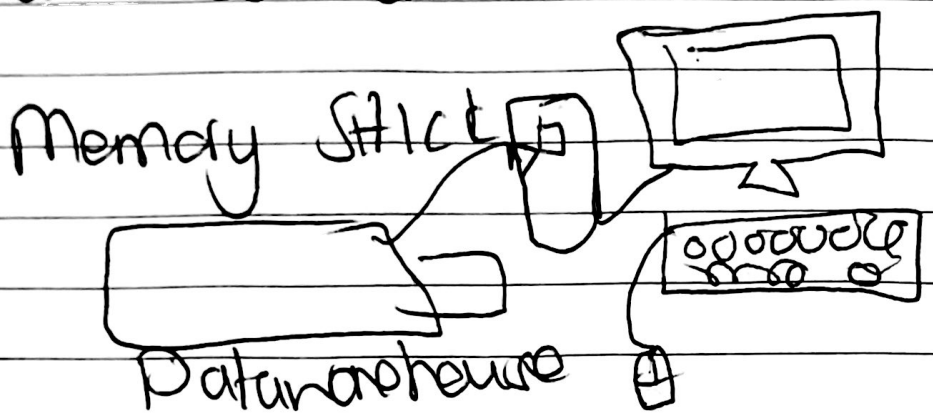
Data Architecture: draws plans of how data works

Source Susem - Shqprile - Capturing
Ingested -

Datawarehouse: Place where we store data so we can drive insight. Centralizedsystems

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Data Management System
A computer system used to manage our data
Allows us to interface our data. Board ~~on~~



connect

Phone — DMS

memory stick — Datawarehouse

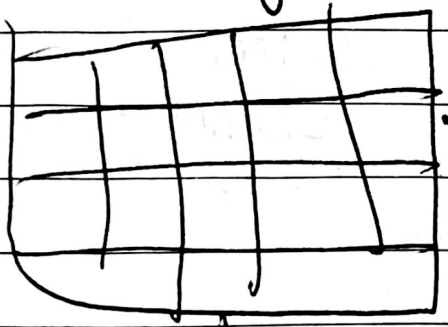
Databrick (Data management System)

Week 2

1. Practice everyday
   30 mins
2. Be consistent regularly
2. Accountability!!

Data

Structured Data



Data stored in a table inside a database

When we have data we need to extract insights from the data

↳ Juice from Data using SQL

Structured Query Language (SQL)

↳ is a scripting language that we use to interact with structured data

We use SQL to retrieve data information from a structured table

student_results

stude id	name	marks
101	Rethi	99
102	Mbali	25
103	Thabo	50
104	Lethabo	60

(1) SELECT: statement (choose)

* All columns
* Allows us to select the specific columns from our table

```
SELECT name,
       marks
FROM students_results
```

name	mark
Rethi	99
Mbali	25
Thabo	50
Lethabo	60

(2) FROM: (specify) statement

* Allows us to specify name of table we are selecting data from
* The select statement allows you to specify the columns you want in your final output

- SELECT * is a way to select all the columns in a table without specifying the column name one by one
- When you say SELECT * you want all the column names in that table
- When selecting specific column names, you separate each column name by a comma (,)

```
SELECT Date,
       Gender,
       Age,
       Quantity

FROM (Table)
```

- On the last column you specify, you don't put the coming

1 SELECT DISTINCT — different usage statement
- We want to get different values from a specific column

- We remove duplicates / we dont repeat the same value.

Syntax: SELECT DISTINCT column_name

FROM table_name;

Fruits_orders

product_id	product_name	Revenue
101	Apple	100
102	Orange	50
103	Apple	60
104	Banana	150

SELECT DISTINT prd_name
FROM Fruits_orders

specify the column name you want diff values from

prd_name
Apple
Orange
Banana

It gives us different values in a specific column.

method 3

```
SELECT  column 1
        column 2
        column 3
FROM    table_name
```

we want to select specific column from our table specify the column we want

Data filtering

- we filter data if we want to get specific information from our data.
- A way to exclude rows of data that do not meet a specific criteria
- filter based on specific info
- filter based on condition

How?

WHERE statement.

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Syntax: SELECT column 1
column 2
column 3
FROM table_name
WHERE column condition;

Students_results

StudentId	name	marks
101	Rothi	99
102	Mbali	25
103	Thabo	50
104	Lethabo	60

Write an SQL query to give me only the names of students that obtained more than 60% in their exam

Syntax:

SELECT *

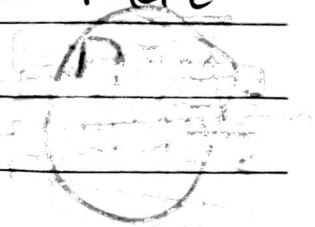
From students_results
WHERE marks > 60;

StudentId	name	marks
Rothi	Rothi	99

Filtered
Reduced
table

Eg. SELECT *
FROM retail_sales_dataset

SELECT *
FROM retail_sales
WHERE Gender = Male



SQL OPERATORS

1. Arithmetic Operators

Operator	Description
+	Addition
-	Substration
*	Multiplication
/	Division

2. Comparison Operators

Operator	Description
=	Equal to
>	Greater than
<	less than
>=	Greater than or Equal to

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<=	Less than or equal to
<>	Not Equal to

3. Logical Operators

Operator	Description
AND	Both conditions must be met
OR	one condition must be met
BETWEEN	must be within the range
IN	must be contained in the list
NOT	Gives the opposite result of what is specified
LIKE	Looks for a pattern

Operator: Tool we can use for different things

Data filtering + WHERE statement

It is a filtering statement that we use to filter our data based on specific conditions

BETWEEN: inclusive statement

- We can use the SQL operator to build the conditions for filtering our data

Syntax: SELECT column 1
column 2
Column 3
FROM table_name
WHERE condition;

Eg.

users

column name

User_id	user-name	City country
1	Nomvula	JHB
2	David	Cape Town
3	Anele	Durban
4	Kabelo	Pretoria
5	Lerato	Port Elizabeth

value

Write an SQL query to give me the name of users in Durban

```
SELECT *
FROM users
WHERE city = 'Durban';
```

condition

we used the operator to build a condition